

Adaptive Energy Regularization for Autonomous Gait Transition and Energy-Efficient Quadruped Locomotion

Boyuan Liang*, Lingfeng Sun*, Xinghao Zhu*, Bike Zhang, Ziyin Xiong, Yixiao Wang, Chenran Li, Koushil Sreenath, Masayoshi Tomizuka

Abstract—In reinforcement learning for legged robot locomotion, crafting effective reward strategies is crucial. Predefined gait patterns and complex reward systems are widely used to stabilize policy training. Drawing from the natural locomotion behaviors of humans and animals, which adapt their gaits to minimize energy consumption, we investigate the impact of incorporating an energy-efficient reward term that prioritizes distance-averaged energy consumption into the reinforcement learning framework. Our findings demonstrate that this simple addition enables quadruped robots to autonomously select appropriate gaits—such as four-beat walking at lower speeds and trotting at higher speeds—without the need for explicit gait regularizations. Furthermore, we provide a guideline for tuning the weight of this energy-efficient reward, facilitating its application in real-world scenarios. The effectiveness of our approach is validated through simulations and on a real Unitree Go1 robot. This research highlights the potential of energy-centric reward functions to simplify and enhance the learning of adaptive and efficient locomotion in quadruped robots. Videos and more details are at <https://sites.google.com/berkeley.edu/efficient-locomotion>

I. INTRODUCTION

Humans and animals exhibit various locomotion behaviors at different speeds, optimizing for their energy efficiency. For instance, humans typically walk at low speeds and run at higher speeds, rarely opting for jumping [1]. Prior research demonstrated through optimal control on planar models the correlation between speed and optimal gait choices concerning the cost of transport (CoT). For quadrupeds, the optimal gaits were four-beat walking¹ at low speeds, trotting at intermediate speeds, and trotting/galloping at high speeds [2].

Due to the rich information in the gaits, using a gait as guidance for locomotion policies is popular among lots of reinforcement learning (RL) based methods [3], [4]. However, crafting a versatile and robust locomotion policy that can adapt to and transition between multiple speeds while generalizing across different platforms poses substantial challenges. One of the main challenges here is the reward design. Gait reference can be used as extended state and extra regularization terms in reward functions to provide more supervision or input for low-level MPC [5] controllers. Previous works [4], [6], [7] trained on different quadruped robots within simulation environments like IsaacGym [8], [9] and successfully transferred these policies to physical

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¹Four-beat walking, two-beat walking, trotting and galloping are typical gaits for quadruped robots defined in [2] based on feet contact schedule.

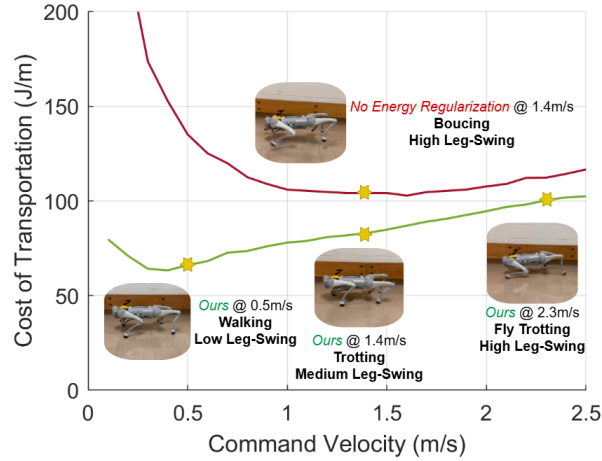


Fig. 1: Compared to the baseline when there is no energy regularization, our single policy (from one-time RL training) autonomously adopted different energy-efficient gaits (walking, trotting and fly trotting). It achieved lower energy consumption (adjusted leg-swing) at varying speeds.

hardware. However, they often necessitate intricate reward designs and weight tuning. Apart from gait information, reward terms like feet-air time and contact force penalizing [6] were also used to encourage specific behaviors and help stabilize the training. While these additional reward components are aimed at inducing or preventing specific behavioral traits, they inadvertently align with the broader objective of reducing energy costs [10], [11]. This convergence prompts a reconsideration of our reward strategy: Could a more straightforward, energy-centric reward term replace the specifically designed terms used in prior policy training and work for general locomotion tasks, including various linear and angular velocity tracking and terrains? Such a term would encapsulate the core objective of reducing energy consumption and fostering stable and efficient locomotion. Gaits at different speeds and terrains are generated posteriorly when the robot finds an energy-efficient way to solve the locomotion task.

Building on the concept that energy-efficient gaits correlate with speed [2] and aligning with prior work emphasizing that energy minimization at pre-selected speed results in the emergence of specific gaits [11], this study investigates a more streamlined reward formulation for energy-efficient locomotion. By focusing on energy minimization without intricately designed reward components, we aim to verify if such a simplified approach can yield stable and effective

自适应能量正则化实现自主步态转换与节能四足运动

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摘要——在腿式机器人运动的强化学习中，设计有效的奖励策略至关重要。预定义步态模式和复杂奖励系统被广泛用于稳定策略训练。受人类和动物根据能量消耗最小化原则自适应调整步态的自然运动行为启发，我们研究了在强化学习框架中引入一个优先考虑距离平均能耗的节能奖励项所带来的影响。我们的研究表明，这一简单的添加项使四足机器人能够自主选择合适的步态——例如在低速时采用四拍行走，在高速时采用小跑——而无需显式的步态正则化。此外，我们提供了调整该节能奖励项权重的指导原则，以促进其在真实场景中的应用。我们的方法通过仿真实验和真实的宇树Go1机器人得到了验证。本研究凸显了以能量为中心的奖励函数在简化和增强四足机器人自适应高效运动学习方面的潜力。视频及更多详情请访问 <https://sites.google.com/berkeley.edu/efficient-locomotion>

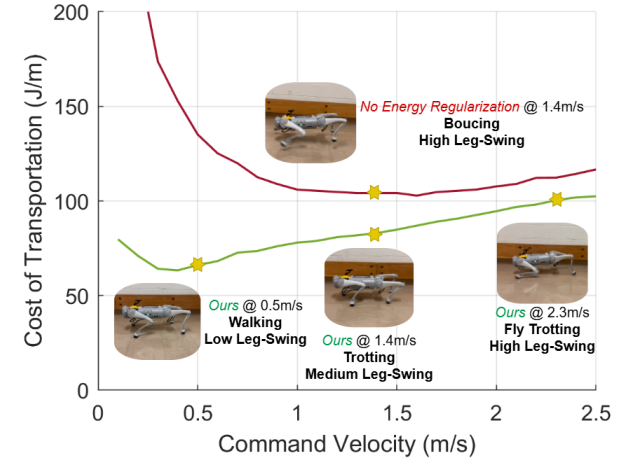


图1: 与没有能量正则化的基线相比，我们的单一策略（通过一次性强化学习训练）自主采用了不同的节能步态（行走、小跑和飞跑）。它在不同速度下实现了更低的能量消耗（调整后的腿部摆动）。

I. 引言

人类和动物在不同速度下展现出多种运动行为，以优化其能量效率。例如，人类通常在低速时行走，在高速时奔跑，很少选择跳跃 [1]。先前的研究通过平面模型上的最优控制证明了速度与最优步态选择之间的相关性，这种相关性涉及运输成本 (CoT)。对于四足动物，最优步态是低速时的四拍行走¹、中速时的小跑，以及高速时的小跑/疾驰 [2]。

由于步态中包含丰富的信息，将步态作为运动策略的引导在众多基于强化学习 (RL) 的方法中十分流行 [3], [4]。然而，设计一种通用且鲁棒的运动策略，使其能够适应多种速度并在不同平台间泛化，同时实现步态转换，这带来了巨大挑战。其中一个主要挑战在于奖励设计。步态参考可以作为扩展状态和奖励函数中的额外正则化项，为底层模型预测控制 (MPC) [5] 控制器提供更多监督或输入。先前的工作 [4], [6], [7] 在 IsaacGym [8][9] 等仿真环境中针对不同的四足机器人进行了训练，并成功将这些策略迁移到物理

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硬件上。然而，这些方法通常需要复杂的奖励设计和权重调优。除了步态信息外，诸如足部离地时间和接触力惩罚 [6] 等奖励项也被用于鼓励特定行为并帮助稳定训练。虽然这些额外的奖励组件旨在诱导或防止特定的行为特征，但它们无意中与降低能量成本 [10], [11] 这一更广泛的目标相一致。这种趋同促使我们重新思考奖励策略：一个更直接、以能量为中心的奖励项，能否取代先前策略训练中使用的那些专门设计的项，并适用于包括各种线速度和角速度跟踪以及地形在内的通用运动任务？这样的项将封装降低能量消耗的核心目标，并促进稳定高效的运动。当机器人找到一种节能的方式来解决运动任务时，不同速度和地形下的步态会随之生成。

基于节能步态与速度 [2]相关这一概念，并借鉴先前强调在预设速度下实现能量最小化会催生特定步态 [11]，的研究，本研究探索了一种更简化的节能运动奖励公式。通过专注于能量最小化而无需精心设计的奖励组件，我们旨在验证这种简化方法能否在四足机器人以不同速度运动时产生稳定且有效的

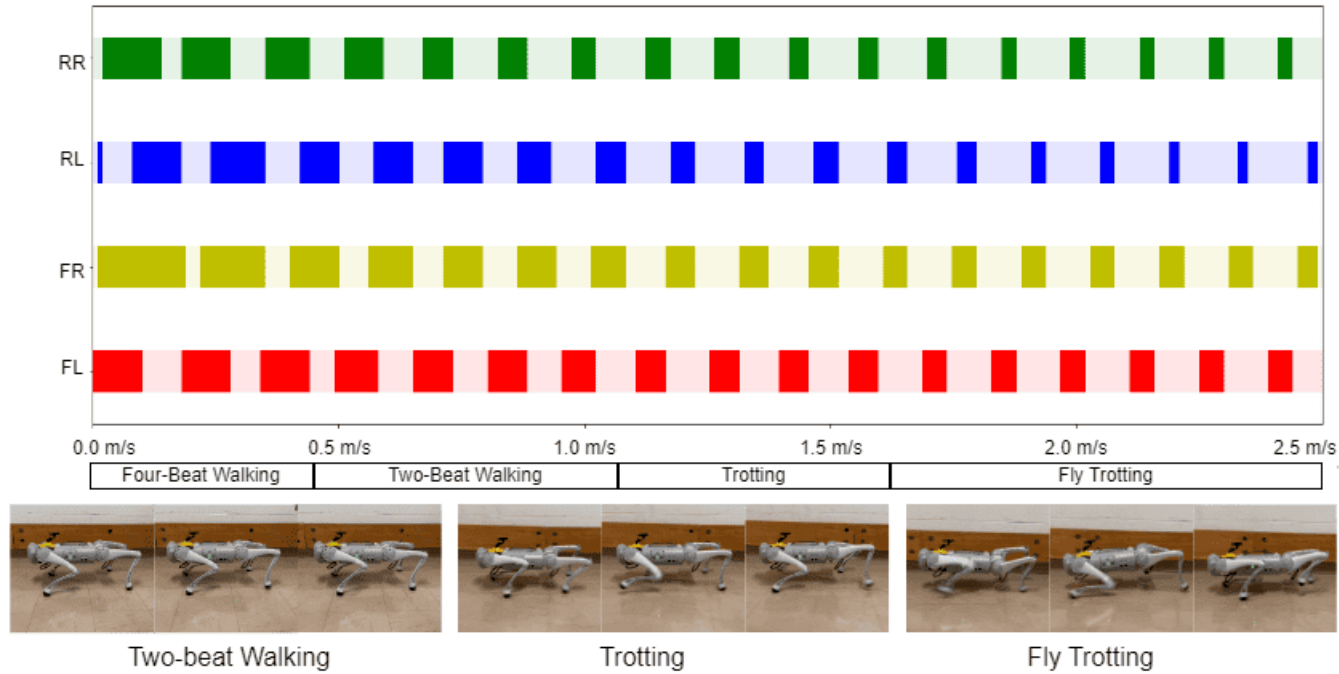


Fig. 2: Gait switching under different command velocities. The policy is generated when $\alpha_{en} = 1.0$. As the command velocity increases, the policy shows automatic gait transition. We also demonstrate snapshots of two beat walking at 0.5 m/s, trotting at 1.4 m/s and fly trotting at 2.3 m/s.

velocity-tracking in quadruped robots across various speeds. Instead of generating multiple velocity-specific energy optimal policies [11], we focus on getting a single energy-optimal policy across all target linear and angular velocities, and different terrains via RL training.

In our research, we investigate the impact of incorporating a distance-averaged energy reward term into the reinforcement learning framework. This reward term directly penalizes energy consumption per unit motion traveled, promoting energy-efficient locomotion across various speeds. We explore the effect of different weightings for this energy reward, observing that both excessively low and high weights can lead to undesirable behaviors such as unnatural movements or immobility. By carefully tuning the weight of the distance-averaged energy reward, we demonstrate its effectiveness in facilitating stable velocity tracking and encouraging the emergence of energy-efficient gaits.

Employing this adaptive reward structure within IsaacGym enables the training of robust policies for the Unitree Go1 [12] quadruped robot. As illustrated in section IV, our methodology identifies appropriate gaits, such as four-beat walking at lower speeds and trotting at higher speeds, without predefined gait knowledge. The energy-efficient policy also shows better performance in circle tracking and terrain clearance tasks. The trained single policy is deployed on a real Go1 robot to verify its stable moving and transition locomotion skills in the real world.

The main contribution of this paper includes:

- Introduction of a streamlined reward formula integrating basic velocity-tracking and distance-averaged energy

minimization to foster stable, velocity-sensitive locomotion policies.

- Demonstration that the derived policies autonomously adopt different energy-efficient gaits at varying speeds without preset gait knowledge.
- Evaluation of the velocity-tracking and energy efficiency across reward structures and weight tunings for Unitree Go1, culminating in the real-world application of these policies on a Go1 robot, affirming their efficacy in stable locomotion and gait transition.

II. RELATED WORKS

A. Reinforcement Learning for Locomotion Skills

After deep RL demonstrated its capability to fit general policies in an unsupervised manner, researchers actively sought its potential to be deployed on legged locomotion. Hwangbo et al. [13] achieved RL-generated walking policies on a plane ground using a pre-trained actuator net to reduce the sim-to-real gap. Further research also achieved training the policy with adaptive actuator net [14] or motor control parameters [15], [16]. In Hwangbo's follow-up works, locomotion on complicated terrains using a similar approach was also accomplished [17], [18], [19]. With modern GPU-accelerated simulators [8], [9], more time-efficient training frameworks were proposed [6], [20]. Due to the intensive reward engineering in these approaches, the model-free RL tends to converge to a single gait, usually trotting gait, which may not be the most efficient for all terrains and target velocities [21].

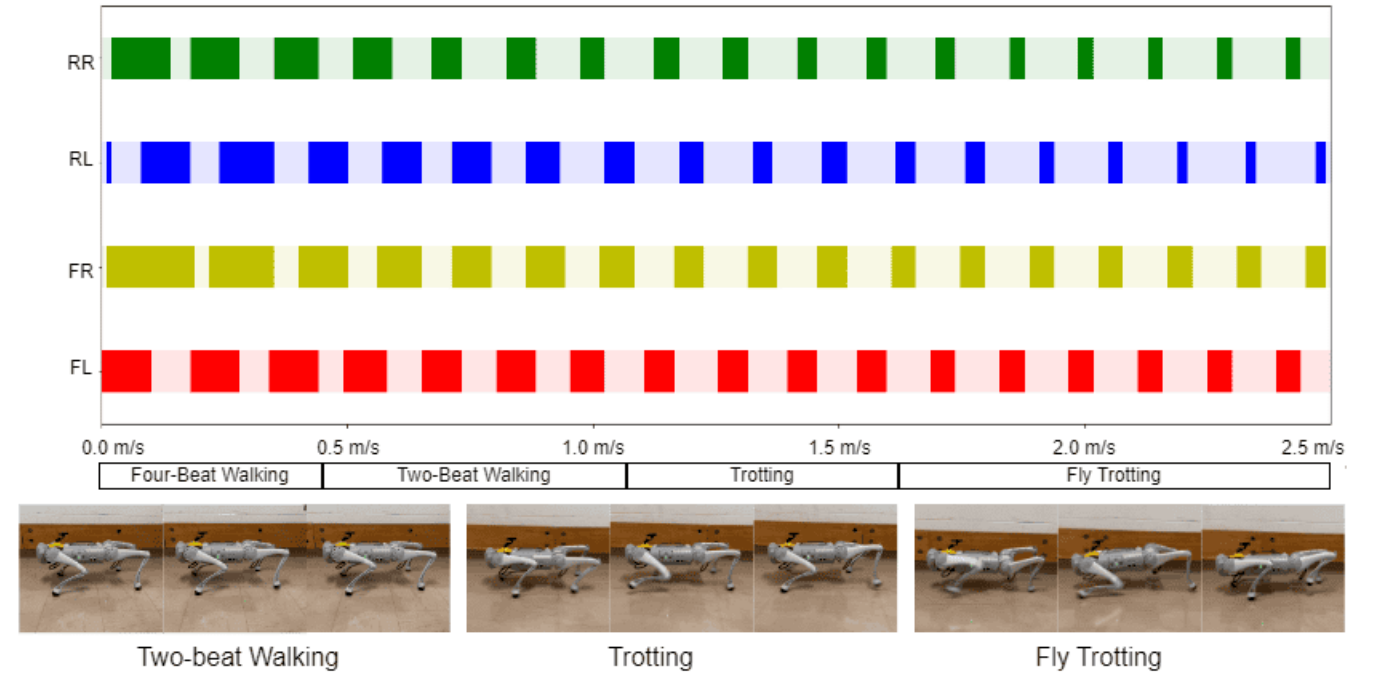


图2: 不同指令速度下的步态切换。该策略在 $\alpha_{en} = 1.0$ 时生成。随着指令速度的增加, 策略展现出自动步态转换。我们还展示了在0.5米/秒时的四拍行走、1.4米/秒时的小跑以及2.3米/秒时的飞跑的截图。

速度跟踪效果。我们并非针对每种速度生成多个特定的能量最优策略 [11], , 而是致力于通过强化学习训练, 获得一个能覆盖所有目标线速度和角速度以及不同地形的单一能量最优策略。

在我们的研究中, 我们探讨了将距离平均能量奖励项纳入强化学习框架的影响。该奖励项直接惩罚单位运动距离的能量消耗, 从而促进在各种速度下的节能运动。我们探索了该能量奖励的不同权重的影响, 观察到权重过低和过高都可能导致不良行为, 例如不自然运动或不动。通过仔细调整距离平均能量奖励的权重, 我们证明了其在促进稳定速度跟踪和鼓励节能步态涌现方面的有效性。

在IsaacGym中采用这种自适应奖励结构, 能够为Unitree Go1[12] 四足机器人训练出鲁棒的策略。如第四节所示, 我们的方法无需预设步态知识即可识别出合适的步态, 例如低速时的四拍行走和高速时的小跑。该节能策略在圆形轨迹跟踪和地形穿越任务中也表现出更好的性能。训练好的单一策略被部署在真实Go1机器人上, 以验证其在现实世界中的稳定运动和过渡运动技能。

本文的主要贡献包括:

- 引入了一种精简的奖励公式, 整合了基本速度跟踪与距离平均能量

最小化, 以促进稳定且对速度敏感的运动策略。

- 证明所推导的策略能够在不同速度下自主采用不同的节能步态, 而无需预设步态知识。
- 针对Unitree Go1, 评估了不同奖励结构和权重调优下的速度跟踪与能效表现, 最终将这些策略实际应用于Go1机器人, 验证了其在稳定运动和步态转换中的有效性。

II. 相关工作

A. 运动技能的强化学习

在深度强化学习展示了其在无监督方式下拟合通用策略的能力后, 研究人员积极寻求将其部署于腿式运动的潜力。Hwangbo等人 [13]利用预训练执行器网络在平面地面上实现了强化学习生成的行走策略, 以减少仿真到现实的差距。进一步的研究还通过自适应执行器网络 [14]或电机控制参数 [15], [16]实现了策略的训练。在Hwangbo的后续工作中, 使用类似方法在复杂地形上的运动也得以实现 [17], [18], [19]。借助现代GPU加速模拟器 [8], [9], , 提出了更高效训练框架 [6], [20]。由于这些方法中密集的奖励工程, 无模型强化学习倾向于收敛到单一步态, 通常是小跑步态, 这可能并非对所有地形和目标速度都是最高效的 [21]。

Many efforts were made to overcome this limitation by promoting the behavioral diversity of legged robots. Researchers in [22] proposed a hierarchical framework that pre-specifies a set of gait primitives and allows an RL model to choose from them. In [5], [23], gait primitives were parameterized using the contact schedule and an RL policy was trained to select these parameters. These works used model predictive control (MPC) as the lower-level controller, which demands preliminary knowledge of locomotion and contact modeling [24]. In [4], behavioral-related arguments such as body height, step frequency, and phase are directly added to the RL model input for end-to-end training. Although various two-beat gaits (where legs touch the ground in pairs) were realized, they cannot generate four-beat gaits (where legs touch the ground in orders) and require manual gait specification in different scenarios. It is more desirable if the quadruped robot can select the most suitable behavior by itself. In [11], a pipeline was proposed to output different gaits under different velocities via minimizing energy consumption. However, this pipeline relies on training and distillation of several velocity-specific RL policies.

B. Energy Studies on Locomotion

The energetic economy of legged robots has always been an important concern for researchers. The cost of transport, namely the amount of energy used per distance traveled, was introduced in [25], [26], where the minimization of CoT was connected to the choice of speeds [27] and step lengths [28] in human locomotion behaviors. The optimal velocity varies for different gaits, allowing animals to transit between different gaits to move at various speeds.

Conceptual legged models of bipedal [29], [30], and quadruped [31], [32], [2] robots are later designed by researchers to search for energetically optimal motions on various gaits. For quadruped robots, genetic algorithms were used in [31] to study potential gaits at different speeds. Our research is mainly inspired by [2], where optimal control problems are formulated on realistic robot models considering the effects of leg mass, plastic collisions, and damping losses. This work uses an unbiased search on energy-efficient locomotion patterns at various velocities and demonstrates the energy vs. velocity curve for different gaits. Inspired by the results, we believe there exists a policy that can transit between gaits at different velocities with an energy-optimization reward. Among previous works, [11] is close to ours in utilizing relations between energy and gaits. It generated multiple policies, with each policy velocity-specific and energy-optimal. In contrast, our work focuses on generating a single energy-optimal policy across various speeds, and it aims to replace the complex-designed reward terms in RL.

III. ENERGY REGULARIZATION

A general form of energy regularized locomotion reward takes the following form:

$$R = (R_{motion} + R_{energy}) * f(R_{aux}), \quad (1)$$

where R_{motion} encourages accurate velocity tracking, R_{energy} discourages energy consumption and R_{aux} includes other necessary rewards to stabilize training. $*$ and f are respectively an arithmetic operator and functions. Common choices are $*$ = +, $f(x) = -x$ [6] and $*$ = $\times$, $f(x) = \exp(-x)$ [4].

In previous work [11], motion rewards include negative squared linear and angular velocity tracking errors; energy rewards include negative time-averaged motor power with a fixed weight $-\tau\dot{q}$; survival bonus is also added. In experiments, we found such a training process unstable across different speeds, primarily for two reasons—1) The negatively unbounded nature of tracking and energy rewards. 2) The scale of energy reward varies across different speeds, and the energy reward weight usually only works within a narrow range of reference speeds. It is hard to find a single energy reward weight value that works for all reference velocities without knowing more simulation settings and training details. As a result, different speeds are trained in different runs separately in [11] to energy-efficient gaits at different speeds.

To overcome this challenge in reward design, we proposed that the energy-related reward function should depend on the robot’s velocity to promote an automatic generation of energy-efficient behavior of legged robots under various reference velocities.

$$R = (R_{motion} + \alpha_{en}R_{en}(v_x, \omega_z)) \exp(-R_{aux}) \quad (2)$$

where v_x is the robot moving velocity. The remaining of this section elaborates on each component in (2).

1) *Motion Rewards*: R_{motion} consists of R_{lin} and R_{ang} , which respectively encourage the legged robot to track the linear reference velocities in two directions $\hat{v}_x, \hat{v}_y$ and angular reference velocities $\hat{\omega}_z$.

$$\begin{aligned} R_{motion} &= R_{lin} + \alpha_{ang}R_{ang}, \\ R_{lin} &= \exp\left(-\frac{|v_x - \hat{v}_x|^2 + |v_y - \hat{v}_y|^2}{\sigma_v}\right), \\ R_{ang} &= \exp\left(-\frac{|\omega_z - \hat{\omega}_z|^2}{\sigma_\omega}\right), \end{aligned} \quad (3)$$

where $\hat{v}_y$ and $\hat{\omega}_z$ are not user-specified commands, but randomly sampled during training as explained in section IV-A. σ_v and σ_ω are scaling factors depending on the training velocity range. The structure of motion rewards, the coefficient $\alpha_{ang} = 0.5$ for angular velocity tracking, and the scaling coefficients follow the default setting in legged-gym [6].

2) *Energy Rewards*: R_{en} rewards the system for consuming less energy while moving.

$$R_{en} = \exp\left(-\frac{\sum_i |\tau_i| |\dot{q}_i|}{\sigma_{en,x}|v_x| + \sigma_{en,z}|\omega_z|}\right) \quad (4)$$

The energy consumption is averaged against the robot’s amount of motion. τ_i ’s are the actuated torque at each joint while $\dot{q}_i$ ’s denotes the joint velocities. $\sigma_{en,x}$ and $\sigma_{en,z}$ are energy scaling constants. Components inside the round bracket of (4) is equivalent to the linear-distance-averaged energy

为克服这一局限，人们做出了许多努力以提升腿式机器人的行为多样性。研究人员在 [22]中提出了一种分层框架，该框架预先指定一组步态基元，并允许强化学习模型从中选择。在 [5], [23], 中，步态基元通过接触时序进行参数化，并训练强化学习策略来选择这些参数。这些工作使用模型预测控制作为底层控制器，这需要具备运动与接触建模的初步知识 [24]。在 [4], 中，行为相关参数（如身体高度、步频和相位）被直接添加到强化学习模型输入中进行端到端训练。尽管实现了多种两拍步态（即双腿成对触地），但无法生成四拍步态（即双腿按顺序触地），且在不同场景下需要手动指定步态。更理想的情况是四足机器人能够自行选择最合适的行为。在 [11],中，提出了一种通过最小化能量消耗在不同速度下输出不同步态的流水线。然而，该流水线依赖于多个速度特定强化学习策略的训练与蒸馏。

B. 运动能量研究

腿式机器人的能量经济性一直是研究人员关注的重点。运输成本，即单位距离所消耗的能量，在 [25], [26], 中被引入，其中运输成本的最小化与人类运动行为中的速度选择 [27]和步长选择 [28]相关联。最优速度因步态而异，这使得动物能够通过切换不同步态来以不同速度移动。

研究人员随后设计了双足 [29], [30]和四足 [31], [32], [2]机器人的概念腿式模型，以探索不同步态下的能量最优运动。对于四足机器人，遗传算法在 [31]中被用于研究不同速度下的潜在步态。我们的研究主要受 [2], 启发，该研究在真实机器人模型上构建了最优控制问题，并考虑了腿部质量、塑性碰撞和阻尼损耗的影响。本工作采用无偏搜索方法，研究不同速度下的节能运动模式，并展示了不同步态的能量-速度曲线。受这些结果的启发，我们认为存在一种策略，能够通过能量优化奖励在不同速度下实现步态切换。在以往的研究中，[11] 与我们利用能量与步态关系的方法最为接近。它生成了多个策略，每个策略针对特定速度且能量最优。相比之下，我们的工作侧重于生成一个跨速度范围的单一能量最优策略，旨在替代强化学习中复杂设计的奖励项。

III. 能量正则化

能量正则化运动奖励的一般形式如下：

$$R = (R_{motion} + R_{energy}) * f(R_{aux}), \quad (1)$$

其中 R_{motion} 鼓励精确的速度跟踪， R_{energy} 抑制能量消耗， R_{aux} 包含其他稳定训练所需的奖励。 $*$ 和 f 分别是算术运算符和函数。常见的选择有 $*$ = +、 $f(x) = -x$ [6] 以及 $*$ = $\times$ 、 $f(x) = \exp(-x)$ [4]。

在先前的工作 [11],中，运动奖励包括负的线性角速度跟踪误差平方项；能量奖励包括带固定权重 $-\tau\dot{q}$ 的负时间平均电机功率；此外还加入了生存奖励。在实验中，我们发现这种训练过程在不同速度下不稳定，主要原因有两个——1)跟踪误差和能量奖励的负无界特性。2) 能量奖励的尺度随速度变化，且能量奖励权重通常仅在较窄的参考速度范围内有效。在不了解更多仿真设置和训练细节的情况下，很难找到一个适用于所有参考速度的单一能量奖励权重值。因此，在 [11] 中，不同速度下的节能步态是通过分别运行不同训练任务获得的。

为了克服奖励设计中的这一挑战，我们提出能量相关奖励函数应依赖于机器人的速度，以促进腿式机器人在各种参考速度下自动生成节能行为。

$$R = (R_{motion} + \alpha_{en}R_{en}(v_x, \omega_z)) \exp(-R_{aux}) \quad (2)$$

其中 v_x 是机器人的移动速度。本节剩余部分将详细阐述公式 (2)中的各个组成部分。

1) *运动奖励*： R_{motion} 由 R_{lin} 和 R_{ang} 组成，它们分别鼓励腿式机器人跟踪两个方向上的线参考速度 $\hat{v}_x, \hat{v}_y$ 以及角参考速度 $\hat{\omega}_z$

$$\begin{aligned} R_{motion} &= R_{lin} + \alpha_{ang}R_{ang}, \\ R_{lin} &= \exp\left(-\frac{|v_x - \hat{v}_x|^2 + |v_y - \hat{v}_y|^2}{\sigma_v}\right), \\ R_{ang} &= \exp\left(-\frac{|\omega_z - \hat{\omega}_z|^2}{\sigma_\omega}\right), \end{aligned} \quad (3)$$

其中 $\hat{v}_y$ 和 $\hat{\omega}_z$ 并非用户指定的指令，而是在训练过程中随机采样的，如第IV-A节所述。 σ_v 和 σ_ω 是取决于训练速度范围的缩放因子。运动奖励的结构、用于角速度跟踪的系数 $\alpha_{ang} = 0.5$ 以及缩放系数均遵循 legged-gym [6]中的默认设置。

2) 能量奖励： R_{en} 用于奖励系统在移动过程中消耗更少的能量。

$$R_{en} = \exp\left(-\frac{\sum_i |\tau_i| |\dot{q}_i|}{\sigma_{en,x}|v_x| + \sigma_{en,z}|\omega_z|}\right) \quad (4)$$

能量消耗根据机器人的运动量进行平均。 τ_i 是每个关节的驱动力矩，而 $\dot{q}_i$ 表示关节角速度。 $\sigma_{en,x}$ 和 $\sigma_{en,z}$ 是能量缩放常数。(4)式圆括号内的分量等价于线性距离平均能量

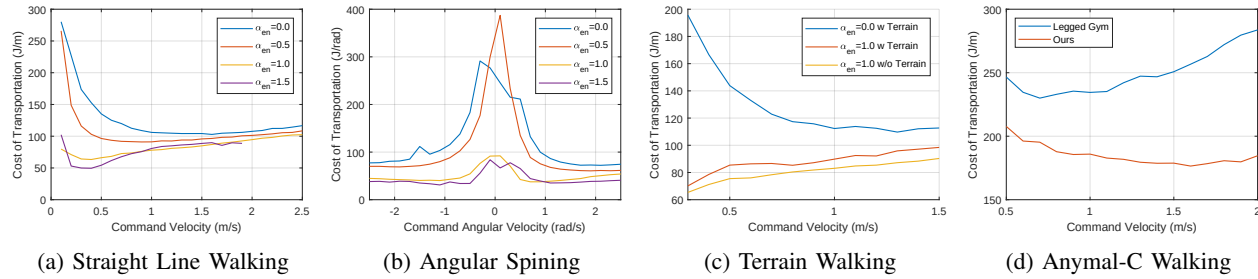


Fig. 3: Ablation study of energy consumption in Unitree Go1 simulation. For straight line walking, reference linear velocities are chosen from 0.1 to 2.5 m/s with 0.1 common gap. The cost of transportation is measured in J/m . For angular spinning, reference angular velocities are chosen from -2.5 to 2.5 rad/s with 0.2 common gap. In both (a) and (b), CoT considerably decreases when α_{en} reaches 1.0. CoT of $\alpha_{en} = 1.5$ when reference velocity of above 1.9 m/s is not plotted because the output velocity drops to zero in this range. This indicates that velocity tracking accuracy will be sacrificed when energy regularization weight α_{en} is too large. For terrain walking, the robot is asked to walk in a straight line on a rough slope terrain with reference linear velocities from 0.3 to 1.5 m/s with 0.1 common gap, because it is hard to walk either too slowly or too quickly on such terrains. (c) shows that the reduced CoT analogously appears on terrains. (d) shows the effect of energy regularization method to ANYmal-C platform with the same parameters $\sigma_{en,x}$, $\sigma_{en,z}$ and α_{en} .

consumption when $\sigma_{en,z} = 0$, which is the conventional definition of the cost of transportation (CoT). Since we are training a linear and angular velocity-dependent policy, we include rotation distance to form a generalized distance while generating the energy reward.

As discussed in [5], directly using the CoT format inside the exponential function in Eq.4 by detecting it from motion reward fails to generate stable policies via end-to-end training. As a result, authors in [5] learn a contact schedule instead and execute it via low-level MPC. In our approach, we deployed the exponential form, which guarantees a positive reward and scale it within $(0, 1]$. We multiply the absolute values of each entry of τ with each entry of $\dot{q}$ and sum them up in (4) to follow the fact that a motor does not get charged back even when the applied torque is opposite to the motion [33]. This adaptive energy regularization term allows us to learn stable locomotion policies across different speeds, tasks, and embodiments.

One may claim that the selection of an appropriate energy reward weight α_{en} in (2) is still nontrivial since a tiny α_{en} diminishes the influence of energy regularization. At the same time, an overly large value can lead to over-enforcement, compromising velocity tracking accuracy. However, the generalized distance design calibrates the energy reward across different speeds, and the exponential function design on motion and energy reward scale both into $(0, 1]$. As a result, a $\alpha_{en} = 1$ works for all cases in our experiments, and the parameter is not that sensitive. This trade-off and detailed ablation study will be shown in Section IV.

3) *Auxiliary Rewards*: Energy regularization alone is usually insufficient for generating proper behaviors. To address this, we add a few auxiliary regularization rewards, including collision avoidance, action rate control and trunk orientation regularization. Higher values of these terms signify less desirable performance, so we deployed a negative exponential function as detailed in (2). Comprehensive information on these auxiliary rewards can be found in the project website.

IV. EXPERIMENTS

The experiments are designed to support the following statements after adding the velocity-dependent energy reward as shown in (4) without specifying gait information.

- The generated RL policy automatically selects an energy efficient action at different command linear and angular velocity.
- The energy reward weight α_{en} should be comparable to motion rewards in order to get a satisfactory RL policy.
- The energy reward can also be used for locomotion training on terrains with minor amendments on the auxiliary reward R_{aux} .

A. Training Setup

We utilized the robot model and PPO training package in [4]. The system outputs the position command of the 12 joints in the next time step. The system inputs include the projected gravity in the robot frame, the commanded x velocities, the commanded yaw rate, each joint's current position and velocity, as well as the action at previous time step. In addition, the inputs of the previous 30 time steps are also given to the RL system. The training episode will reset after 1000 time steps or if any part of the robot except its feet touches the floor.

1) *Details of Reward Parameters*: In motion rewards (3), both σ_v and σ_ω were fixed at 0.25. In energy rewards (4), the energy scaling constants are fixed at $\sigma_{en,x} = 1000$ and $\sigma_{en,z} = 500$. As stated in section III, we found that the energy reward R_{en} alone is insufficient to regularize Go1's behavior, which is likely due to the lighter weight compared to its motor power. Thus, following the settings in [4], we add the fixed auxiliary reward R_{aux} . This auxiliary reward is derived mainly from safety concerns, such as penalizing limb-ground collision, out-of-range joint position, and high frequency joint action. The details of R_{aux} can be found on the project website. Compared to [4], we did not include any gait-related rewards.

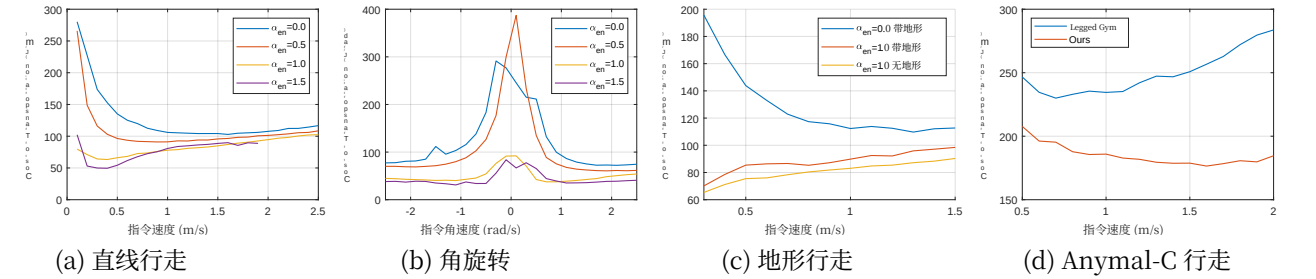


图 3: 宇树 Go1 仿真中能量消耗的消融研究。对于直线行走, 参考线速度从 0.1 到 2.5 m/s 选取, 间隔为 0.1。运输成本以 J/m 为单位测量。对于角旋转, 参考角速度从 -2.5 到 2.5 rad/s 选取, 间隔为 0.2。在 (a) 和 (b) 中, 当 α_{en} 达到 1.0 时, 运输成本显著降低。当参考速度高于 1.9 m/s 时, $\alpha_{en} = 1.5$ 的运输成本未绘制, 因为在此范围内输出速度降至零。这表明当能量正则化权重 α_{en} 过大时, 速度跟踪精度会受到影响。对于地形行走, 机器人被要求在粗糙斜坡地形上直线行走, 参考线速度从 0.3 到 1.5 m/s 选取, 间隔为 0.1, 因为在此类地形上行走过慢或过快都很困难。(c) 显示运输成本的降低同样出现在地形上。(d) 展示了能量正则化方法对 ANYmal-C 平台的影响, 参数为 $\sigma_{en,x}$, $\sigma_{en,z}$ 和 α_{en} 。

消耗, 当 $\sigma_{en,z} = 0$ 时, 这是运输成本 (CoT) 的传统定义。由于我们正在训练一个线速度和角速度相关策略, 因此我们在生成能量奖励时纳入旋转距离以形成广义距离。

如 [5] 所述, 直接从运动奖励中检测运输成本, 并在公式(4)的指数函数内使用其格式, 无法通过端到端训练生成稳定策略。因此, [5] 的作者转而学习接触时序, 并通过底层模型预测控制执行。在我们的方法中, 我们采用了指数形式, 这保证了正奖励, 并将其缩放至 $(0, 1]$ 范围内。我们将 τ 每个分量的绝对值与 $\dot{q}$ 每个分量相乘, 并在(4)中求和, 以遵循以下事实: 即使施加的扭矩与运动方向相反 [33], 电机也不会反向充电。这种自适应能量正则化项使我们能够学习跨不同速度、任务和实体的稳定运动策略。

有人可能会认为, 在(2)中选择合适的能量奖励权重 α_{en} 仍然不简单, 因为过小的 α_{en} 会削弱能量正则化的影响。同时, 过大的值可能导致过度约束, 从而影响速度跟踪精度。然而, 广义距离设计校准了不同速度下的能量奖励, 而指数函数设计将运动奖励和能量奖励都缩放至 $(0, 1]$ 范围内。因此, 在我们的实验中, 一个 $\alpha_{en} = 1$ 适用于所有情况, 且该参数并不敏感。这种权衡和详细的消融研究将在第四节中展示。

3) *辅助奖励*: 仅靠能量正则化通常不足以生成合理的行为。为解决这一问题, 我们添加了几项辅助正则化奖励, 包括避碰、动作速率控制和躯干朝向正则化。这些项的值越高, 表示性能越不理想, 因此我们采用了如(2)中详述的负指数函数。关于这些辅助奖励的完整信息, 可在项目网站上找到。

IV. 实验

实验旨在支持以下论断: 在未指定步态信息的情况下, 添加如(4)所示的与速度相关的能量奖励后, 所得结果。

- 生成的强化学习策略能够在不同指令线速度和角速度下自动选择节能动作。
- 能量奖励权重 α_{en} 应与运动奖励相当, 才能获得满意的强化学习策略。
- 能量奖励也可用于地形上的运动训练, 只需对辅助奖励 R_{aux} 进行小幅修改。

A. 训练设置

我们使用了 [4] 中的机器人模型和 PPO 训练包。系统输出下一时间步 12 个关节的位置指令。系统输入包括机器人坐标系中的投影重力、指令 x 方向速度、指令偏航角速率、每个关节的当前位置和速度, 以及上一时间步动作。此外, 前 30 个时间步的输入也会提供给强化学习系统。训练回合将在 1000 个时间步后或机器人除脚部以外的任何部位接触地面时重置。

1) *奖励参数详情*: 在运动奖励 (3) 中, σ_v 和 σ_ω 均固定为 0.25。在能量奖励 (4) 中, 能量缩放常数固定为 $\sigma_{en,x} = 1000$ 和 $\sigma_{en,z} = 500$ 。如第三节所述, 我们发现仅凭能量奖励 R_{en} 不足以规范 Go1 的行为, 这很可能是因为其重量相对于电机功率较轻。因此, 遵循 [4], 中的设置, 我们添加了固定的辅助奖励 R_{aux} 。该辅助奖励主要源于安全考虑, 例如惩罚肢体-地面碰撞、关节位置超限以及高频关节动作。 R_{aux} 的详细信息可在项目网站上找到。与 [4], 相比, 我们没有包含任何步态相关奖励。

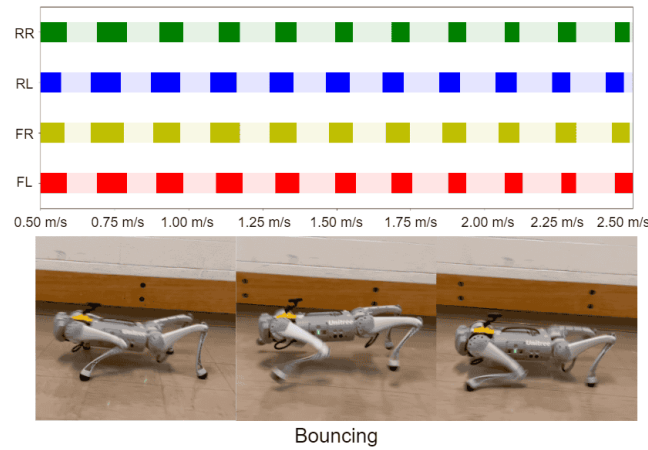


Fig. 4: Gait under different command velocities when $\alpha_{en} = 0.0$. The policy shows bouncing gait across all command velocities, which is not an energy-efficient choice of gait. We also demonstrate snapshots of bouncing gait, which is the four legs touches and leaves the ground almost simultaneously.

2) *Curriculum and Domain Randomization*: We deployed curriculum technique on command velocities. The sampling range of linear and angular velocities start at $[-1, 1]$ m/s and $[-1, 1]$ rad/s. The sampling range increases when the total reward achieves a certain threshold, and the maximal sampling range is set at $[-5, 5]$ m/s and $[-5, 5]$ rad/s. The ground coefficient of friction is randomized between $[0.05, 1.5]$. We also added a disturbance to the robot mass with a uniform random value in $[-0.1, 3.0]$ kg. A uniformly distributed noise was added to the observation. All these randomized domain parameters are renewed every time the training episode resets, except observation noise is resampled after every time step.

3) *Amendments for Training on Terrains*: All previous techniques are deployed on flat ground trainings. We also tested adding energy regularization while training on terrains. The training terrain shape is rough slope adapted from [6].

B. Translation and Rotation

To demonstrate natural emergence of efficient locomotion gaits, we evaluate the trained walking policy under different reference velocities. Fig. 2 demonstrates a trial run where the legged robot was commanded to move from $\hat{v}_x = 0.0$, to 2.5 m/s. The robot accelerates at $0.5m/s^2$. We plot the gait recorded from the Go1 real-world deployment. We can see that our trained policy exhibits four-beat walking at low speed (around 0.0 to 0.4 m/s), where four legs touches the ground in front-left, front-right, rear-left, rear-right sequence. Then, the robot shows two-beat walking (around 0.4 to 1.1 m/s) where the four legs touch the ground in diagonal pairs and present observable moments where four legs touches the ground at the same time. At medium speed (around 1.1 to 1.7 m/s), the policy exhibits an trotting gait. At this gait, the four legs still touch the ground in diagonal pairs, but there

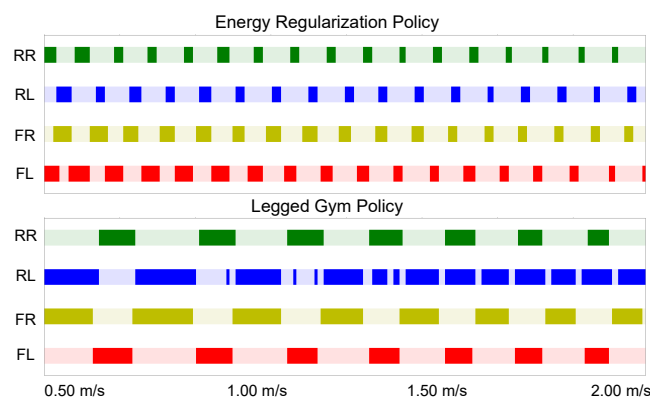


Fig. 5: Gait comparison of ANYmal-C between energy regularization and original legged gym policy [6]. Similar gait transition from walking to trotting also appears when energy regularization is applied. In the original legged gym policy, the lifting height of rear right leg is very low, so it has several unexpected mild contacts with the ground. Videos of ANYmal-C simulation can be found on our project website.

is neither noticeable moment where the four legs touch the ground at the same time, nor moment where all four legs are in the air. At high speed (around 1.7 m/s and beyond), the trained policy exhibits a fly-trotting gait, which is similar to trotting gait, except there are observable moments where all four legs are in the air. This gait transition is endorsed by previous works [2], [5] that walking and trotting are respectively the most energy-efficient gaits under low and high speeds.

The result in Fig. 2 is generated with $\alpha_{en} = 1.0$. In Fig. 3a and Fig. 3b, we compare CoT and linear velocity tracking results across different α_{en} . When weight is small (like 0.0 or 0.5), the effect of energy regularization is minor, so the generated policy has a much higher CoT. Fig. 4 shows that when $\alpha_{en} = 0.0$, the robot exhibits a bouncing gait across the whole velocity domain, which is not efficient [2].

The proposed energy regularization can be similarly applied to other quadruped platforms. Experiments were conducted on the ANYmal-C simulation environment [6] with the same scaling constants $\sigma_{en,x} = 1000$ and $\sigma_{en,z} = 500$ as well as the regularization weight $\alpha_{en} = 1.0$. The generated policy similarly showed preferred gait transition from walking to trotting, which is show in Fig. 5. As shown in Fig. 3d, it also successfully reduced CoT in comparison to the original legged gym settings [6].

C. Circle Tracking

We also tested the circle-tracking capacity of the policy. We ask the robot to walk with linear velocity 1.0 m/s and angular velocity 0.5 rad/s. This is equivalent to asking the robot to follow a circle with radius 2 m. We moved the robot in a semicircle with predefined center in simulation and recorded its position at each time step. Assume the distance between the robot and the circle center is $d(t)$ at time step t ,

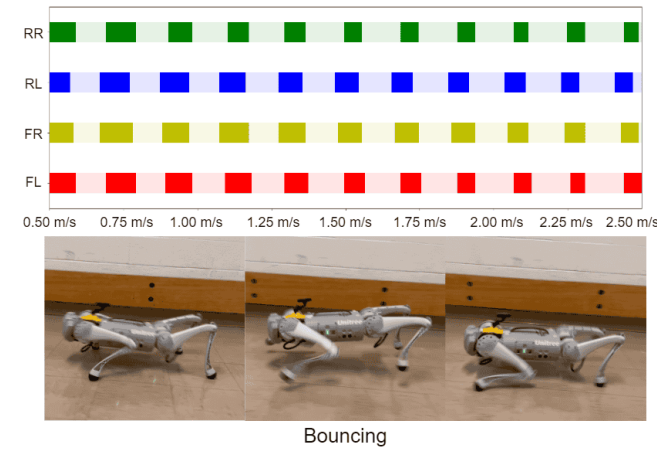


图4: 当 $\alpha_{en} = 0.0$ 时不同指令速度下的步态。该策略在所有指令速度下均表现为弹跳步态，这不是一种节能的步态选择。我们还展示了弹跳步态的截图，即四条腿几乎同时离地触地。

2) 课程学习与域随机化：我们对指令速度采用了课程学习技术。线速度和角速度的采样范围起始于 $[-1, 1]$ 米/秒和 $[-1, 1]$ 弧度/秒。当总奖励达到特定阈值时，采样范围会增加，最大采样范围设置为 $[-5, 5]$ 米/秒和 $[-5, 5]$ 弧度/秒。地面摩擦系数在 $[0.05, 1.5]$ 之间随机化。我们还对机器人质量添加了扰动，扰动值为 $[-0.1, 3.0]$ 千克内的均匀随机值。观测值中添加了均匀分布的噪声。所有这些随机化的域参数在每次训练回合重置时更新，但观测噪声在每个时间步后重新采样。

3) 地形训练的改进措施：前述所有技术均应用于平坦地面训练。我们还测试了在地形训练中引入能量正则化的效果。训练地形形状为粗糙斜坡，改编自 [6]。

B. 平移与旋转

为了展示高效运动步态的自然涌现，我们评估了训练后的行走策略在不同参考速度下的表现。图2展示了一次试验运行，其中腿式机器人被指令从 $v_x = 0.0$ 移动到 2.5米/秒。机器人在 $0.5m/s^2$ 处加速。我们绘制了从 Go1 实际部署中记录的步态。可以看出，我们的训练策略在低速（约0.0至0.4米/秒）时表现出四拍行走，即四条腿按左前、右前、左后、右后的顺序触地。随后，机器人表现出两拍行走（约0.4至1.1米/秒），此时四条腿以对角对形式触地，并出现四条腿同时触地的可观察时刻。在中速（约1.1至1.7米/秒）时，策略表现出小跑步态。在此步态下，四条腿仍以对角对形式触地，但

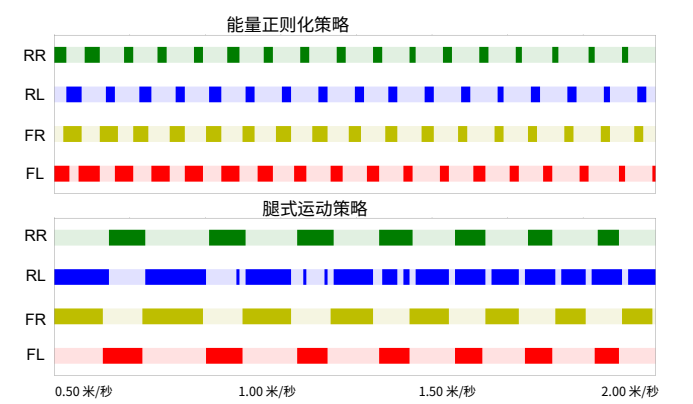


图5: ANYmal-C在能量正则化与原始腿式运动策略 [6] 下的步态对比。应用能量正则化后，同样出现了从行走向小跑的步态转换。在原始腿式运动策略中，右后腿的抬腿高度极低，导致其与地面发生多次意外的轻微接触。ANYmal-C的仿真视频可在我们的项目网站上查看。

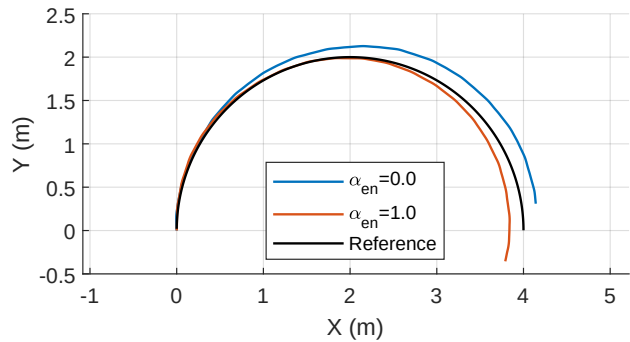
既没有四条腿同时触地的明显时刻，也没有所有四条腿同时腾空的时刻。在高速（约1.7米/秒及以上）时，训练后的策略表现出腾空小跑步态，该步态与小跑步态相似，但存在所有四条腿同时腾空的可观察时刻。这种步态转换得到了先前研究 [2], [5] 的支持，即行走和小跑分别是低速和高速下最节能的步态。

图2中的结果是在 $\alpha_{en} = 1.0$ 下生成的。在图3a和图3b中，我们比较了不同 α_{en} 下的运输成本和线速度跟踪结果。当权重较小时（如0.0或0.5），能量正则化的效果较小，因此生成的策略具有更高的运输成本。图4显示，当 $\alpha_{en} = 0.0$ 时，机器人在整个速度域内表现出弹跳步态，这种步态效率不高 [2]。

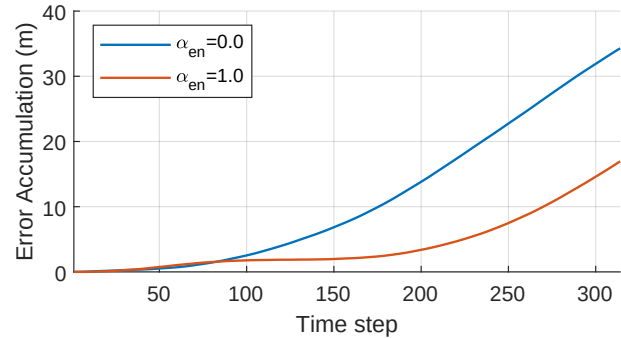
所提出的能量正则化方法可以类似地应用于其他四足平台。实验在ANYmal-C仿真环境 [6] 中进行，使用了相同的缩放常数 $\sigma_{en,x} = 1000$ 和 $\sigma_{en,z} = 500$ 以及正则化权重 $\alpha_{en} = 1.0$ 。生成的策略同样显示出从行走到小跑的优选步态转换，如图5所示。如图3d所示，与原始的腿式运动设置 [6] 相比，该方法也成功降低了运输成本。

C. 圆形轨迹跟踪

我们还测试了该策略的圆形跟踪能力。我们要求机器人以线速度1.0米/秒和角速度0.5弧度/秒行走。这相当于要求机器人沿着半径为2米的圆形路径运动。我们在仿真中将机器人移动到一个具有预定义中心的半圆上，并记录每个时间步的位置。假设在时间步 t 时，机器人与圆心之间的距离为 $d(t)$,



(a) Semicircle Tracking Sketch Plot



(b) Semicircle Tracking Error Accumulation

Fig. 6: Open-loop semicircle tracking comparison between $\alpha_{en} = 0.0$ and $\alpha_{en} = 1.0$. Since energy regularization reduces random motions irrelevant to target motions, the tracking error for $\alpha_{en} = 1.0$ accumulates considerably slower than $\alpha_{en} = 0.0$.

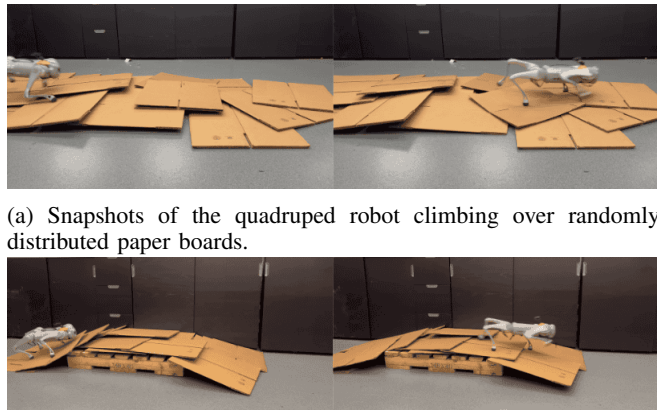
the error $e(t)$ is defined as the absolute value of the difference between d and the 2 m pre-defined radius, $|d(t) - 2|$. We compare the accumulated error $\sum_{i=1}^T e(t)$ between $\alpha_{en} = 0.0$ and 1.0 policies.

The results are demonstrated in Fig. 6. The trajectory radius of $\alpha_{en} = 1.0$ policy is closer to the 2 m target and the error also accumulates much slower. This is because that energy regularization can effectively reduce motions that are not directly related to the target 1.0 m/s, 0.5 rad/s motion.

D. Terrain Clearance

The effect of energy regularization on quadruped gait can also be deployed on terrain clearance. Since it is hard to walk either very fast or very slow on a rough slope, the training velocity range is limited to only $[-1.5, 1.5]$ m/s. When no energy regularization presents, the quadruped robot also tends to show a bouncing gait. When $\alpha_{en} = 1.0$, a more natural trotting gait appears. As indicated in Fig. 3, the policy generated from $\alpha_{en} = 1.0$ is also a more energy efficient gait.

We tested the trained policy on the real Go1 quadruped robot. Fig. 7 shows experiment snapshots. The robot successfully cleared a 20 cm step covered by paper boards, which is a considerable high step compared to the size of Go1.



(a) Snapshots of the quadruped robot climbing over randomly distributed paper boards.
(b) Snapshots of the quadruped robot climbing over a 20 cm step covered by paper boards.

Fig. 7: Quadruped robot clearing terrains.

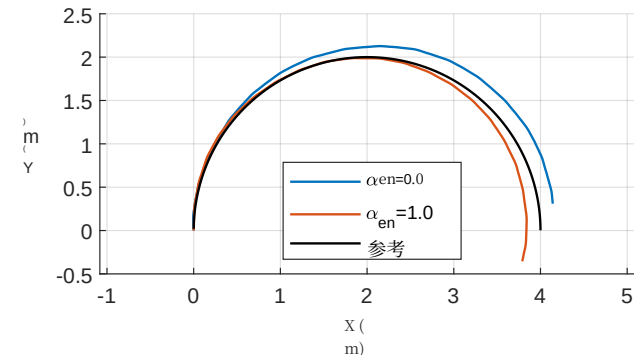
V. DISCUSSION AND CONCLUSION

This paper presented a novel approach to energy-efficient locomotion in quadruped robots by implementing a simplified, energy-centric reward strategy within a reinforcement learning framework. Our method demonstrated that quadruped robots, specifically Unitree Go1, could autonomously develop and transition between various gaits across different linear and angular velocities without relying on predefined gait patterns or intricate reward designs. The adaptive energy reward function, adjusted based on velocity, enabled these robots to select the most energy-efficient locomotion strategies naturally.

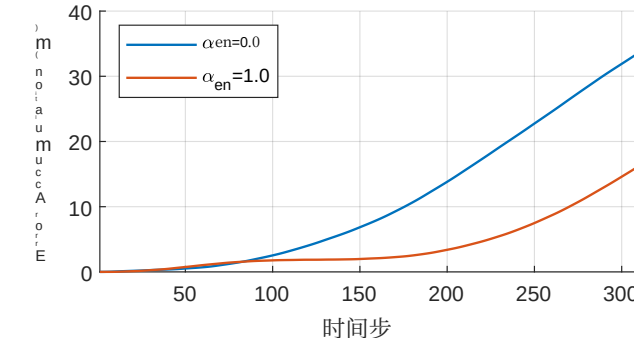
There are more extended locomotion tasks that we can make use of the energy-centric training, where the motion reward is no longer velocity tracking but includes more task-specific rewards. This would require the system to dynamically adjust its strategy in response to different task rewards. While this study concentrated on locomotion tasks, the underlying principle of leveraging energy efficiency to drive behavior selection holds broader potential. Future work could explore applying this energy-centric approach across different robotic tasks. For instance, manipulation and interaction tasks could also benefit from strategies prioritizing energy efficiency, potentially finding natural, efficient behaviors analogous to those observed in biological systems [34], [35]. Such a framework would align robotic systems more closely with sustainability principles and environmental consciousness.

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(a) 半圆跟踪示意图



(b) 半圆跟踪误差累积

图6: $\alpha_{en} = 0.0$ 与 $\alpha_{en} = 1.0$ 之间的开环半圆跟踪比较。由于能量正则化减少了与目标运动无关的随机运动, $\alpha_{en} = 1.0$ 的跟踪误差累积速度明显慢于 $\alpha_{en} = 0.0$ 。

误差 $e(t)$ 定义为 d 与 2 米预定义半径之差的绝对值, 即 $|d(t) - 2|$ 。我们比较了 $\alpha_{en} = 0.0$ 和 1.0 策略之间的累积误差 $\sum_{i=1}^T e(t)$ 。

结果如图6所示。 $\alpha_{en} = 1.0$ 策略的轨迹半径更接近 2 米的目标, 且误差累积速度也慢得多。这是因为能量正则化能有效减少与目标 1.0 米/秒、0.5 弧度/秒运动不直接相关的运动。

D. 地形净空

能量正则化对四足步态的影响同样可应用于地形净空。由于在粗糙斜坡上无论是快速还是慢速行走都较为困难, 训练速度范围被限制在仅 $[-1.5, 1.5]$ 米/秒。当不存在能量正则化时, 四足机器人也倾向于表现出弹跳步态。当 $\alpha_{en} = 1.0$ 时, 则会出现更自然的小跑步态。如图3所示, 由 $\alpha_{en} = 1.0$ 生成的策略也是一种能量效率更高的步态。

我们在真实的 Go1 四足机器人上测试了训练好的策略。图 7 展示了实验快照。机器人成功越过了由纸板覆盖的 20 厘米台阶, 相对于 Go1 的尺寸而言, 这是一个相当高的台阶。



(a) 四足机器人攀爬随机分布的纸板快照。

(b) 四足机器人攀爬覆盖纸板的 20 厘米台阶快照。

图7: 四足机器人清理地形。

V. 讨论与结论

本文提出了一种新颖的方法, 通过在强化学习框架中实施简化的、以能量为中心的奖励策略, 实现四足机器人的节能运动。我们的方法证明, 四足机器人 (特别是 Unitree Go1) 能够在不同线速度和角速度下自主发展并在各种步态之间转换, 而无需依赖预定义步态模式或复杂的奖励设计。基于速度调整的自适应能量奖励函数, 使这些机器人能够自然地选择最节能的运动策略。

我们可以利用以能量为中心的训练来执行更多扩展的运动任务, 其中运动奖励不再是速度跟踪, 而是包含更多任务特定奖励。这将要求系统根据不同的任务奖励动态调整其策略。虽然本研究专注于运动任务, 但利用能量效率驱动行为选择的基本原则具有更广泛的潜力。未来的工作可以探索将这种以能量为中心的方法应用于不同的机器人任务。例如, 操作与交互任务也可以从优先考虑能量效率的策略中受益, 从而可能找到类似于生物系统[34], [35]中观察到的自然、高效的行为。这样的框架将使机器人系统更符合可持续性原则和环境意识。

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